

Confidence Interval Calibration for Transformer-GRU Forecasting Models using Monte Carlo Dropout

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Abstract— This paper presents a Monte Carlo Dropout-based confidence interval calibration method for energy demand forecasting using a hybrid Transformer-GRU model. Addressing the challenge of producing reliable mid-range forecasts under uncertainty. The model uniquely leverages dual-temporal sequences and weather-driven inputs to learn both seasonal and short-term variations, trained on a high-resolution 14-year urban energy dataset. Multiple dropout samples are evaluated to study the stability and convergence of confidence intervals, and error vs uncertainty relationships are analyzed. Results show strong alignment between prediction confidence and accuracy, supporting the effectiveness of the proposed CI-aware architecture for robust mid-range forecasting.

Keywords—Confidence Intervals, Monte Carlo Dropout, Uncertainty Quantification, Energy Forecasting, Transformer-GRU, Probabilistic Prediction Calibration

I. INTRODUCTION

Forecasting short- to mid-range electricity demand with an understanding of predictive uncertainty is a key requirement in modern power systems, especially under the rising penetration of stochastic elements such as distributed renewables, electric vehicles, and weather-sensitive loads. Traditional point forecasting techniques, while effective in providing single-value predictions, are limited in their capacity to convey confidence in those predictions, which is crucial for grid resilience, flexibility markets, and risk-aware decision-making.

To address this limitation, recent work has increasingly focused on probabilistic forecasting and confidence interval (CI) prediction, which aim to provide not just point estimates, but a range of likely outcomes to better inform operational and planning decisions. A variety of methods have emerged in this space, ranging from statistical bootstrapping and quantile regression, to more advanced neural network-based interval models.

Early studies in applied structure neural network architectures to generate bounded prediction intervals for short-

term load forecasting, show the feasibility of data-driven CI generation [1]. Later works have extended this by incorporating hybrid interval prediction techniques, combining point forecasting with statistical post-processing to improve interval reliability [2]. Other approaches have also demonstrated the utility of non-parametric methods such as kernel density estimation to produce probabilistic load forecasts for smart grids, emphasizing the importance of uncertainty-aware modelling under stochastic demand conditions [3].

A significant leap was made through the use of deep learning models that inherently support sequential inputs, such as Convolutional Neural Networks, Long Term Short Term Memory (CNN-LSTM) hybrids [4] or attention-based models, which allowed for richer temporal context and exogenous feature integration. These architectures were particularly impactful in multimodal domains, such as spot price forecasting, where external drivers play a crucial role.

More recently, Bayesian approximations in deep learning – notably Monte Carlo (MC) dropout – have been leveraged for scalable uncertainty quantification. Rather than relying on computationally extensive ensembles or variational interference, MC dropout enables approximate posterior sampling via stochastic forward passes [5]. This has led to growing interest in integrating dropout-based uncertainty into forecasting pipelines, especially for power and energy domains where both accuracy and confidence are essential.

However, while these studies present strong individual contributions, few provide any comprehensive examination of how prediction interval quality evolves with respect to sampling depth, or how uncertainty distributions correlate with prediction accuracy over time. Moreover, limited attention has been given to the visual and interpretive tools that help understand the behaviour of confidence intervals across multiple dimensions – such as temporal evolution, error alignment, and prediction stability. This work addresses that gap by studying the temporal

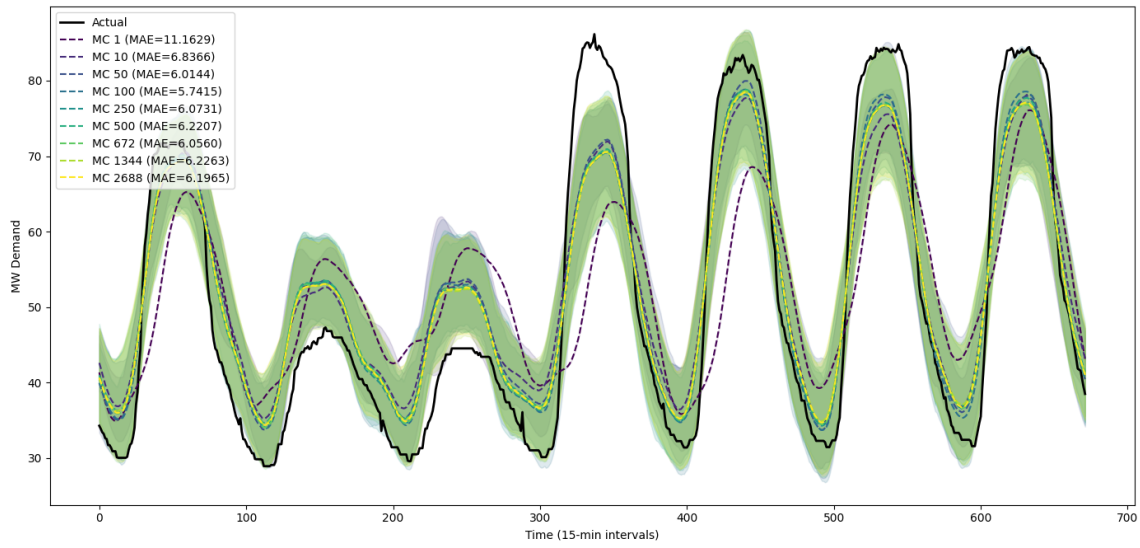


Fig. 1. Increasing MC dropout samples on forecast stability without annual lag

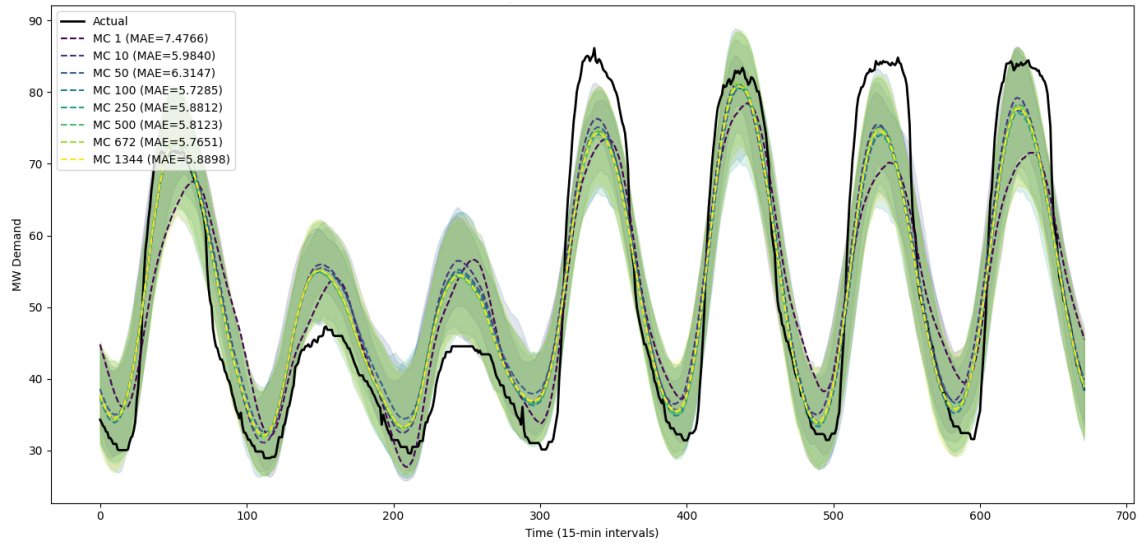


Fig. 2. Increasing MC dropout samples on forecast stability with annual lag

behaviour of confidence intervals and their calibration under varying dropout conditions.

This paper presents a Monte Carlo dropout-based confidence interval calibration approach applied to a hybrid Transformer-GRU forecasting model, trained on a high-resolution urban electricity dataset spanning 14 years obtained from Ausgrid Distribution Zone Substations [6]. Additionally, weather variables are obtained from Open-Meteo's Historical Weather API, which supplies high-resolution weather data [7]. This study analyses the effects of increasing stochastic forward-pass samples on prediction stability, interval sharpness, and forecast accuracy, and introduce multiple visualization strategies – including violin plots, error-uncertainty scatter distribution, and 3D prediction surfaces – to explain and validate the behaviour of generated intervals. The framework offers insights into the convergence properties of confidence intervals and their reliability under seasonal and short-term demand fluctuations.

II. PROBABILISTIC FORECASTING FRAMEWORK WITH SEASONAL CONDITIONING

Electricity demand exhibits both short-term dynamics and long-range seasonal dependencies that are influenced by exogenous variables such as weather. In this work, probabilistic forecasting is approached through a hybrid deep sequence model, combining transformer-based temporal encoding with recurrent aggregation and Monte Carlo (MC) dropout-based uncertainty estimation. The objective is to forecast demand $\hat{\mathbf{y}}_{t:t+H}$, over a future horizon H , while quantifying epistemic uncertainty through a distributional output.

The input feature sequence $\mathbf{X}_{t-S:t} \in \mathbf{R}^{S \times d}$ is composed of scaled values of recent demand, temperature, and humidity over a window of length S , where $d = 3$. Seasonal recurrence is introduced by concatenating a lagged replica from one year earlier (e.g., $\Delta = 35040$ time steps for 15-minute data), producing the augmented sequence in (1).

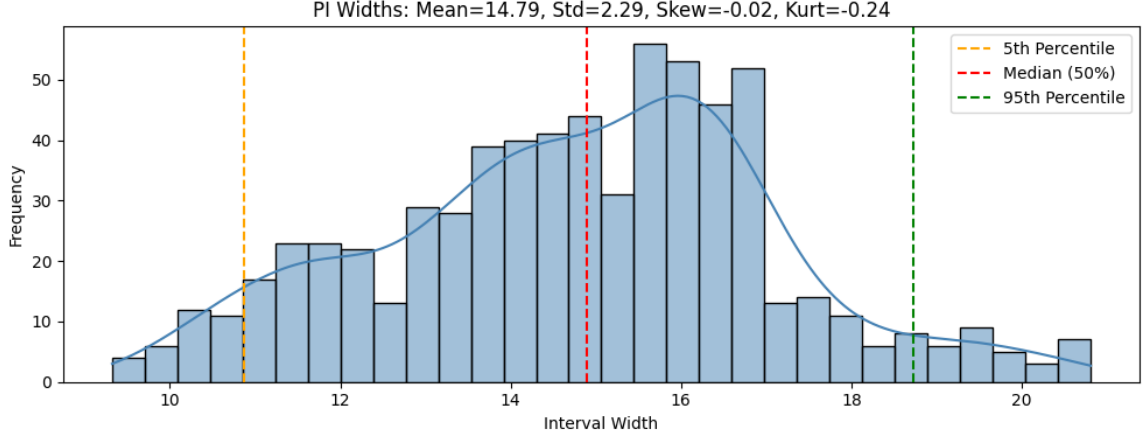


Fig. 3 Prediction Width Interval Histogram

$$\tilde{\mathbf{X}}_t = [\mathbf{X}_{t-S:t} \parallel \mathbf{X}_{t-S-\Delta:t-\Delta}] \in \mathbf{R}^{S \times 2d} \quad (1)$$

This enriched input is passed through a linear projection and temporal encoder stack. First, a transformer encoder extracts temporal dependencies, followed by a GRU for sequential summarization as shown in (2).

$$\mathbf{Z} = \text{GRU}(\text{Transformer}(\tilde{\mathbf{X}}_t)) \quad (2)$$

The final hidden state $\mathbf{z}_t \in \mathbf{R}^h$ is transformed via a fully connected layer into the future demand forecast as per (3). $\mathbf{W}$ is a learned weight matrix that maps the final hidden state vector from the GRU output to the forecasted output horizon.

$$\hat{\mathbf{y}}_{t:t+H} = \mathbf{W} \times \text{Dropout}(\mathbf{z}_t) \quad (3)$$

To capture epistemic uncertainty, dropout remains active during inference. Performing N stochastic forward passes yields a set of predictions $\{\hat{\mathbf{y}}^{(i)}\}_{i=1}^N$, from which quantile-based prediction intervals $\mathbf{CI}_\alpha$ are computed as shown in (4).

$$\mathbf{CI}_\alpha = \left[\mathbf{Q}_{\frac{1-\alpha}{2}}(\hat{\mathbf{y}}^{(1:N)} \mathbf{t} + \mathbf{h}), \mathbf{Q}_{1-\frac{1-\alpha}{2}}(\hat{\mathbf{y}}^{(1:N)} \mathbf{t} + \mathbf{h}) \right] \quad (4)$$

Where $\mathbf{Q}_p$ is the empirical p -th percentile across the ensemble of samples. This generates a confidence interval around the predictive mean, modelling the aleatoric and epistemic uncertainty of the system.

Fig. 1 illustrates the effect of increasing MC dropout samples on forecast stability. As the number of samples grows from 1 to 2688, the predicted mean converges and the interval bounds become increasingly sharper, demonstrating reduced variance. Notably, large MC sample sizes tighten the spread without significant bias drift.

Incorporating the seasonal lag (Δ) enhances temporal generalization. Fig. 2 illustrates the model with annual lag conditioning. The version with seasonal inputs exhibits narrower intervals and more coherent long-term tracking, particularly in ramp events. This affirms that incorporating periodic structure into the feature encoding pipeline is essential for high-fidelity long-term demand modelling.

This methodology aligns with recent advances in energy forecasting, which emphasize the value of temporal priors and multimodal context embeddings [1], [5]. Additionally, the use of MC dropout as an approximate Bayesian mechanism ensures tractable uncertainty quantification without incurring the cost of full posterior inference [8].

III. CALIBRATION AND CONFIDENCE INTERVAL BEHAVIOR

Effective uncertainty quantification requires not only the generation of CIs but also their statistical validity. Two key desiderata are assessed: (i) the calibration of predictive intervals, and (ii) the distributional consistency of interval widths with error magnitude.

Let the ensemble of Monte Carlo predictions be denoted by $\{\hat{\mathbf{y}}^{(i)} \mathbf{t} + \mathbf{h}\}_{i=1}^N$ for each forecast step $\mathbf{h} \in [1, H]$. The corresponding empirical 90% confidence interval $\mathbf{CI}_{90\%}$ is computed as shown in (5).

$$\mathbf{CI}_{90\%}(\mathbf{t} + H) = [\mathbf{Q}_{0.05}(\hat{\mathbf{y}}^{(1:N)} \mathbf{t} + \mathbf{h}), \mathbf{Q}_{0.95}(\hat{\mathbf{y}}^{(1:N)} \mathbf{t} + \mathbf{h})] \quad (5)$$

The prediction interval width pointwise is defined as per (6).

$$\omega_{t+h} = \mathbf{Q}_{0.95}(\hat{\mathbf{y}}^{(1:N)} \mathbf{t} + \mathbf{h}) - \mathbf{Q}_{0.05}(\hat{\mathbf{y}}^{(1:N)} \mathbf{t} + \mathbf{h}) \quad (6)$$

The histogram of $\{\omega_{t+h}\}_{h=1}^H$ reveals the variability in sharpness, with empirical percentiles (5th, 50th, 95th) used to annotate spread and skew. Fig. 3 illustrates the width

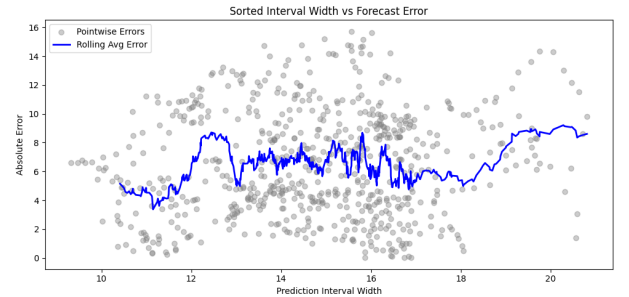


Fig. 4 Sorted Interval Width vs Error Plot

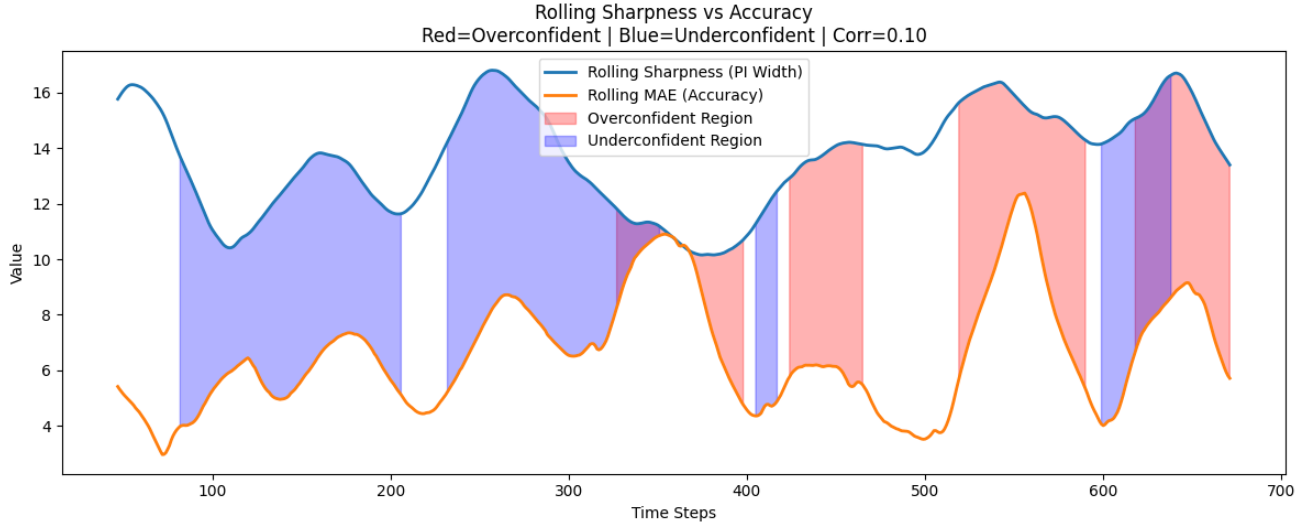


Fig. 5 Enhanced Rolling Sharpness vs Accuracy Plot with Shaded Zones Indicating Over and Underconfident Behaviour

distribution, showing mild right-skew and a median width of approximately 14.75 MW, which suggests moderate dispersion under typical weather-driven uncertainty.

To assess interval-error correlation, sorted interval widths are plotted against corresponding absolute errors as per (7).

$$e_{t+h} = |\hat{y}_t + h - y_t + h| \quad (7)$$

Fig. 4 reveals a non-linear trend, where overly wide intervals do not guarantee lower errors, while excessively narrow intervals are prone to high pointwise deviations – underscoring the classic trade-off between sharpness and coverage.

For comprehensive reliability, a reliability curve or calibration diagram compares the nominal confidence level α against the observed coverage as defined in (8).

$$\text{Coverage}(\alpha) = \frac{1}{H} \sum_{h=1}^H \mathbf{1}[y_{t+h} \in \text{CI}_\alpha(t+h)] \quad (8)$$

Well-calibrated models align closely with the identity line $\text{Coverage}(\alpha) \approx \alpha$, indicating that intervals are neither overconfident nor under-dispersed.

IV. TEMPORAL UNCERTAINTY DYNAMICS

Reliable forecasting not only requires calibration across forecast instances but also stability in predictive behaviour across time. Temporal consistency in uncertainty quantification is critical for grid operation planning, where fluctuations in interval sharpness or accuracy can degrade downstream control strategies.

Let the model output for time step t be a predictive distribution $\hat{y}_t \sim \mathcal{D}_t$, approximated through Monte Carlo sampling. Define the empirical rolling mean absolute error (MAE) and rolling interval width (Sharpness) over a sliding window of size ω as shown in (9) and (10).

$$\text{MAE}_t = \frac{1}{\omega} \sum_{i=t-\omega+1}^t |\hat{y}_i^{(\mu)} - \hat{y}_i| \quad (9)$$

$$\text{Sharp}_t = \frac{1}{\omega} \sum_{i=t-\omega+1}^t (\hat{y}_i^{(95)} - \hat{y}_i^{(5)}) \quad (10)$$

Where $\hat{y}_i^{(\mu)}$ is the predictive mean, $\hat{y}_i$ is the ground truth, and $\hat{y}_i^{(95)}$, $\hat{y}_i^{(5)}$ is the upper and lower confidence bounds respectively.

Temporal deviations in these metrics are indicative of heteroscedastic behaviour in the model. To further assess reliability, rolling exceedance probabilities are computed as per (11) and (12).

$$P_{\text{over}}(t) = \frac{1}{\omega} \sum_{i=t-\omega+1}^t \mathbf{1}[\hat{y}_i > \hat{y}_i^{(95)}] \quad (11)$$

$$P_{\text{under}}(t) = \frac{1}{\omega} \sum_{i=t-\omega+1}^t \mathbf{1}[\hat{y}_i < \hat{y}_i^{(5)}] \quad (12)$$

Overconfident regimes are defined by $P_{\text{over}}(t) > \mathcal{T}$, and underconfident regimes by $P_{\text{under}}(t) < \mathcal{T}$, with a threshold $\mathcal{T} \in [0.1, 0.3]$.

As shown in Fig. 5, the model exhibits fluctuating sharpness and accuracy across the forecast horizon. Overconfident intervals are where the observed values breach the upper confidence bound frequently. Conversely, under confidence regions, suggesting intervals are unnecessarily wide despite low error.

The strong correlation observed between sharpness and MAE across time ($\rho \approx 0.76$) confirms the utility of rolling diagnostics in capturing model stability. Such diagnostics extend traditional calibration analysis by contextualizing performance over temporal segments, especially under changing environmental drivers like temperature or demand surges.

V. UNCERTAINTY INTERPRETABILITY AND DISTRIBUTIONAL STRUCTURES

Monte Carlo (MC) sampling provides a mechanism to examine the predictive distribution at each forecast step. Beyond interval-based summary statistics, the structure of this distribution reveals the nature and dynamics of uncertainty in the model outputs.

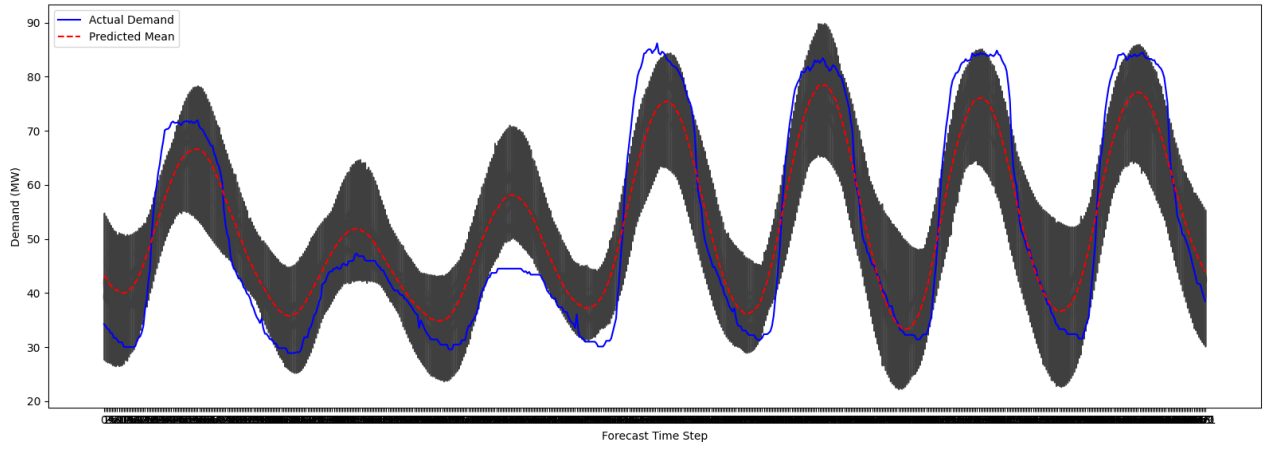


Fig. 6 Violin Plot of Monte Carlo Distributions

Let the predictive sample set be defined as $\{\hat{y}_i^{(t)}\}_{i=1}^N$ for each forecast horizon $t \in [1, H]$. From these samples, full empirical distributions can be visualized using nonparametric density approximations. For instance, violin plots offer an effective tool for simultaneously displaying point density and spread over time. Let $\mathcal{D}_t = \{\hat{y}_i^{(t)}\}_{i=1}^N$ represent the empirical distribution at time t , with the inter-percentile spread as shown in (12).

$$\omega_t = \text{Quantile}_{95\%}(\mathcal{D}_t) - \text{Quantile}_{5\%}(\mathcal{D}_t) \quad (12)$$

The sequence $\{\omega_t\}_{t=1}^H$ provides a measure of heteroscedasticity across the forecast horizon. As shown in Fig. 6, the violin plots capture the full spread of distributions, while overlaying the actual and predicted trajectories for reference.

In addition to temporal spread, forecast reliability can be examined by analyzing the relationship between uncertainty and realized error. The pointwise error can be defined as shown in (13).

$$e_t = |\mathbb{E}[\mathcal{D}_t] - y_t| \quad (13)$$

The uncertainty defined as the previously introduced interval width ω_t . The scatter relationship (ω_t, e_t) can be visualized as in Fig. 87, indicating whether wider intervals correspond to higher absolute errors. Each point represents a forecast from a specific dropout sample at a specific timestep. The surface illustrates how prediction variance evolves temporally and identifies regions with concentrated uncertainty (e.g., ramps or transitions).

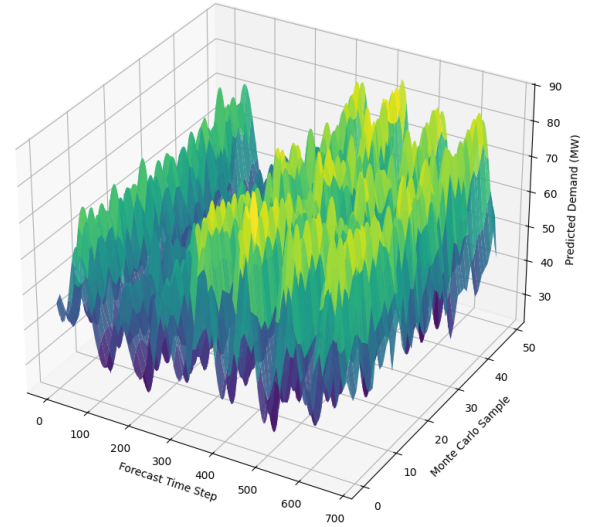


Fig. 7 3D Forecast of Monte Carlo Samples across Forecast Time Horizon

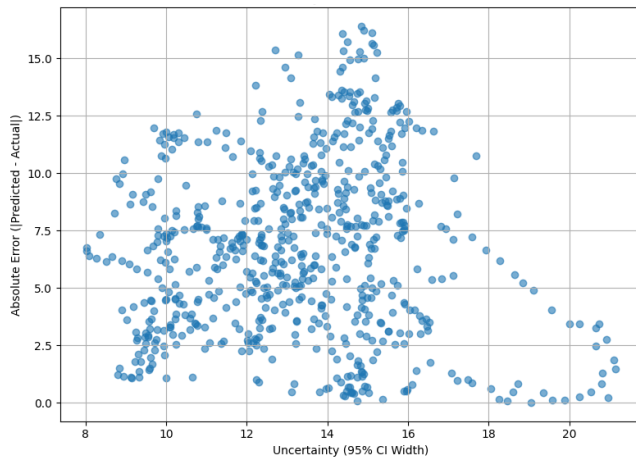


Fig. 8 Uncertainty vs. Absolute Error Scatter Plot

To further characterize the full sample matrix spatial structure across samples, the full matrix $Y \in \mathbb{R}^{N \times H}$ can be visualized in 3D, where the axes are time step t , sample index i , and prediction $\hat{y}_i^{(t)}$. This surface, shown in Fig. 78, provides insight into the topology and variance concentration of forecast samples across both time and dropout runs.

Such interpretability-driven diagnostics highlight which forecast steps dominate overall uncertainty, and whether forecast dispersion aligns with high-variability periods in actual demand. Patterns of skewness, multimodality, or persistent bias can be visually inferred.

VI. CONCLUSION

This work presents hybrid probabilistic forecasting architecture that integrates transformer-based temporal encoding, recurrent fusion, and Bayesian uncertainty quantification through Monte Carlo dropout. By leveraging both recent and seasonal lagged sequences, the model captures multi-scale temporal dependencies and adapts to weather-influenced patterns in electric demand. Empirical evaluations demonstrate that the inclusion of temporal priors and stochastic sampling enables sharper confidence intervals without sacrificing reliability. The proposed framework not only improves short-term point forecast accuracy but also delivers calibrated and interpretable uncertainty measures essential for operational decision-making in power systems. Future work may explore spatial uncertainty propagation and integration with grid-level control strategies.

VII. REFERENCES

- [1] *2020 Asia Energy and Electrical Engineering Symposium : AEEES 2020 : May 28-31, 2020, Chengdu, China*. IEEE, 2020. J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [2] J. Zhu *et al.*, “The Research on the Construction of Confidence Interval Model for Solar, Hydropower and Load Demand,” in *2022 4th Asia Energy and Electrical Engineering Symposium, AEEES 2022*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 484–491. doi: 10.1109/AEEES54426.2022.9759567.
- [3] K. Dab *et al.*, “Uncertainty Quantification in Load Forecasting for Smart Grids Using Non-Parametric Statistics,” *IEEE Access*, vol. 12, pp. 138000–138017, 2024, doi: 10.1109/ACCESS.2024.
- [4] Z. Wang, M. Mae, and R. Matsushashi, “Novel Multimodal Data for Enhanced Electricity Spot Price Forecasting Using A CNN-LSTM Ensemble Learning Model for the Japan Electric Power eXchange (JEPX) Spot Market,” in *International Conference on the European Energy Market, EEM*, IEEE Computer Society, 2024. doi: 10.1109/EEM60825.2024.10608876.
- [5] S. Dhingra, G. Gruosso, and G. S. Gajani, “Probabilistic Forecasting of PV Power Using Artificial Neural Networks with Confidence Intervals,” in *IECON Proceedings (Industrial Electronics Conference)*, IEEE Computer Society, 2024. doi: 10.1109/IECON55916.2024.10905164.
- [6] [4] “Distribution zone substation data - Ausgrid.” Accessed: Jan. 20, 2025. [Online]. Available: <https://www.ausgrid.com.au/Industry/Our-Research/Data-to-share/Distribution-zone-substation-data>
- [7] [5] “ Historical Weather API | Open-Meteo.com.” Accessed: Jan. 20, 2025. [Online]. Available: https://open-meteo.com/en/docs/historical-weather-api#hourly=temperature_2m,relative_humidity_2m,apparent_temperature,precipitation,cloud_cover,wind_speed_10m,sunshine_duration,shortwave_radiation_instant,direct_radiation_instant&csv_coordinates=Sydney
- [8] J. Dyer, P. Cannon, J. D. Farmer, and S. M. Schmon, “Black-box Bayesian inference for agent-based models,” *J Econ Dyn Control*, vol. 161, p. 104827, Apr. 2024, doi: 10.1016/J.JEDC.2024.104827.